

# DREPADetect: An Optimized YOLOv8 Model for High-Precision Detection of Sick Cell

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**Abstract**— The detection of blood cells in patients with sickle cell disease represents a significant challenge in computer vision, primarily due to the cells' small size and tendency to overlap. In this study, we propose DREPADetect, an optimized detection model for blood cells, based on a modified YOLOv8 architecture. To enhance accuracy for small objects, we streamlined the model by selectively removing blocks and layers dedicated to detecting medium and large-sized objects. This adaptation allows computational resources to focus on the morphological features specific to blood cells. We optimized the backbone and neck components of the model by reinforcing blocks specialized in high-resolution feature extraction, which is crucial for detecting small objects. This approach enables DREPADetect to minimize the influence of artifacts and off-target elements in images, focusing attention on regions of interest (ROIs) corresponding to blood cells. Experimental results demonstrate that DREPADetect achieves superior performance compared to state-of-the-art models, with a precision of 87.6%, a recall of 87.6%, and a mean average precision (mAP@50) of 90.3% on a dedicated dataset of blood cells from sickle cell patients. These results outperform reference models such as YOLOv8 and derivatives. This study highlights the effectiveness of DREPADetect for the accurate detection of blood cells in a complex clinical context, offering promising prospects for automated medical image analysis in sickle cell disease.

**Keywords**— YOLOv8, Blood, Diagnosis, Detection, Sickle Cell Disease.

## I. INTRODUCTION

Sickle cell disease (SCD) is a group of hereditary disorders caused by a mutation in the beta-globin gene, where glutamic acid is replaced by valine at the sixth codon [1]. This mutation causes red blood cells to polymerize into rigid, sickle-shaped forms, impairing blood flow and leading to vaso-occlusion, stroke, acute chest syndrome, pulmonary hypertension, and leg ulcers [2].

SCD affects around 300,000 newborns annually, with over 80% of cases in Africa [3]. In Senegal, 10% of the population carries the gene, and 1,700 babies are born with SCD yearly. Projections estimate 14.2 million affected births by 2050 [4].

Affordable, reliable diagnostic tools are vital, particularly in low-resource settings. Traditional blood smear analysis is limited by skilled labor dependency, manual errors, delays, and lack of advanced equipment [5]. Conventional methods (e.g., Watershed, Hough Transform) fail with overlapping cells and real-time integration, driving the adoption of deep learning models (CNNs, Faster R-CNN, YOLO) [6]. Yet, computational constraints hinder real-time deployment.

To address these challenges, we propose DREPADetect, an optimized YOLOv8 model for sickle cell detection. Enhanced for small object detection, it focuses on distinctive features in blood smear images. Tested on datasets with varied cell configurations (including overlaps), our model outperforms YOLOv8 in accuracy and efficiency.

The key contributions of our study are following; (1) Enhanced YOLOv8 for sickle cell detection, improving accuracy in complex diagnostic cases; (2) Optimized inference time, meeting real-time clinical constraints; (3) Developed 'SKDrepaData', an annotated dataset (100× magnification) to mitigate data scarcity; (4) Refined overlapping cell detection, ensuring precise morphological analysis; (5) Maintains high accuracy, even under imaging artifacts, resolution inconsistencies, and lighting variations; (6) Validated on multiple datasets ('SKDrepaData' and 'Erythrocyte IBD1'), proving robustness in low-resource environments.

The remainder of this paper is structured as follows: Section II reviews related work. Section III details DREPADetect's optimized architecture. Section IV covers experiments, followed by results and discussions in Section V. Section VI concludes with future directions.

## II. RELATED WORK

Research on hematological disease diagnosis has advanced significantly, with AI-based approaches enhancing sickle cell detection through machine learning and deep learning models. Early methods relied on image segmentation and feature extraction but faced challenges in accuracy, overlapping cell detection, and real-time feasibility [7].

Traditional techniques, such as Otsu’s thresholding and the Watershed algorithm, were used for feature extraction [8]. While effective, they suffered from high error rates with overlapping cells and dependence on handcrafted features [9], [10], [11]. Methods like edge detection (Sobel filter) and circular Hough transform (CHT) struggled with noise and complex cell structures[10], [11], prompting the shift to deep learning [12].

88.0%, demonstrating YOLOv8’s efficiency in hematological analysis.

Despite advancements, existing models still face challenges with overlapping cells, computational efficiency, and real-time deployment. To address these limitations, we introduce DREPADetect, an optimized YOLOv8-based model tailored for sickle cell detection. By focusing on small

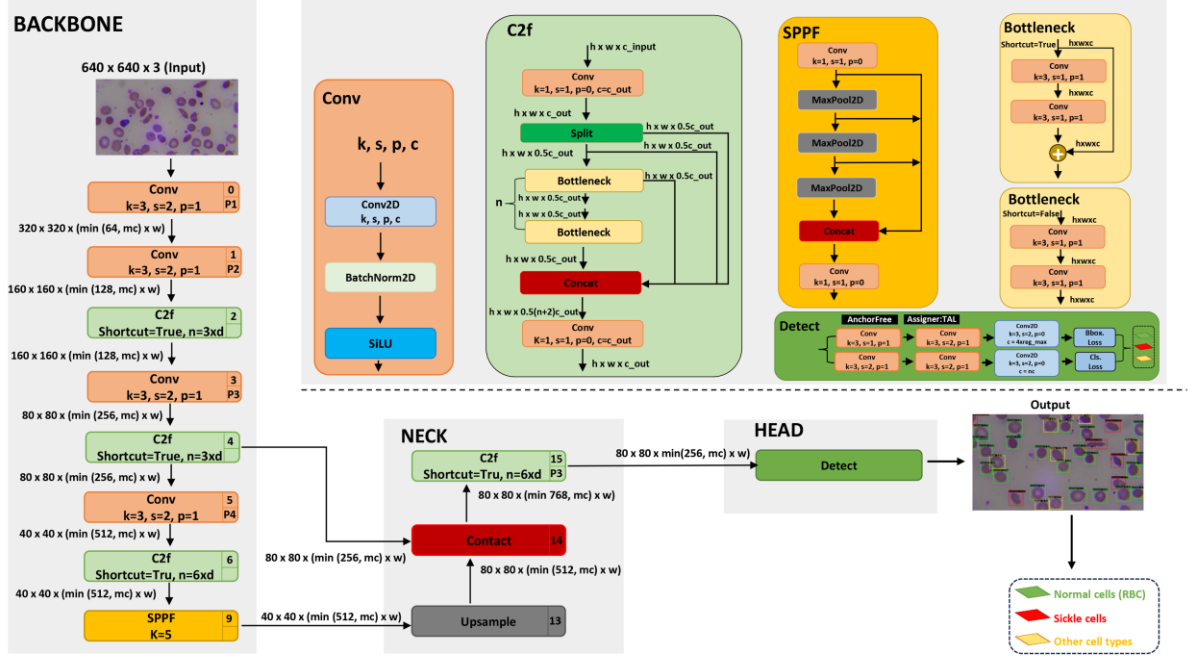


Fig 1. An optimized proposed Architecture for DREPADetect.

CNN-based models have gained prominence. Bhatia et al.[13] developed a custom CNN trained on 10,377 single-cell images, outperforming LightGBM, Inception V3, ResNet, and VGG. However, traditional CNNs remain computationally intensive and require large datasets for optimal performance.

More recent works [14], [15], [16], [17] have leveraged YOLO and R-CNN-based object detection architectures for blood cell classification. John Paolo et al. [18] applied YOLOv3 for sickle cell diagnosis, achieving a 50% confidence score on a small dataset, highlighting YOLO’s potential but also its dataset dependency. Karunasena et al. [19] developed an R-CNN-based model, integrating CNN feature extraction and SVM classification, reaching 90% accuracy but with high computational costs unsuited for real-time practice on low-end CPUs.

Faster R-CNN has been applied to anemia-related cell detection, with Haq et al. [20] achieving 91.83% accuracy for RBCs, 97.90% for WBCs, and 88.36% for platelets. However, it remains limited by inference delay, thus unsuited for real-time clinical use.

Recent studies have optimized YOLO-based models for sickle cell detection. Wang et al. [21] used YOLOv7 with adaptive preprocessing, achieving mAP@50 scores of 80.2% and 81.9%. Zeyu et al. [22] propose YOLO-BC, an improved YOLOv8 variant, outperforming YOLOv8 with 80.9% accuracy, mAP@50 of 90.1, F1-score of 85%, and Recall of

object detection, reducing computational costs, and improving real-time inference, DREPADetect maintains high accuracy while overcoming key drawbacks of prior models.

DREPADetect leverages elementary morphological structures, extracting features based on minimal cell diameter to effectively handle overlapping cells and feature extraction challenges. It outperforms state-of-the-art models by enhancing inference speed, enabling real-time decision-making, and addressing previous limitations making DREPADetect an efficient, accurate, and real-time diagnostic tool for SCD detection.

### III. PROPOSED METHODOLOGY

Existing sickle cell detection methods struggle with real-time deployment and over-lapping erythrocyte identification; DREPADetect overcomes these challenges.

By exploiting the dimensional uniformity of erythrocytes (~7–8  $\mu\text{m}$ ) and the fine details captured at high magnification ( $\times 40$ ,  $\times 100$ ), DREPADetect accurately identifies subtle morphological features of sickle cells. High magnification imaging reveals intricate cell structures, essential for precise falciform shape detection.

Thus, the key challenge lies in designing a deep learning model that focuses on these critical features while optimizing both detection accuracy and processing speed. The proposed

architecture (Fig 1.) enhances inference time and detection precision by prioritizing the most relevant morphological aspects.

#### A. DREPADetect Architecture

The DREPADetect architecture consists of three components: Backbone, Neck, and Head. The Backbone, as the core innovation for feature extraction, is the core innovation of this study. In YOLOv8, the Backbone processes  $640 \times 640 \times 3$  input images through convolutional layers, reducing spatial resolution while extracting features. Given the dimensional consistency of sickle cells ( $\sim 7\text{--}8\mu\text{m}$ ) and the irrelevance of large receptive fields for detecting individual cells [23], our key improvement lies in removing unnecessary detection blocks.

Rather than focusing on medium and large object detection, as in YOLOv8, DREPADetect retains only blocks optimized for small object detection, making it better suited for erythrocyte identification. Specifically, Conv 0, Conv 1, Conv 3, Conv 5, and C2f blocks are preserved, while Blocks 7-8, 10-12, and 16-21—which target medium and large objects—are removed.

Additionally, the Spatial Pyramid Pooling Fast (SPPF) module in the Backbone and the Contact and Upsample blocks in the Neck are retained to enhance feature extraction, fusion, and reconstruction, improving accuracy and efficiency. The Neck refines extracted features, which the Head then classifies and localizes. By streamlining YOLOv8, DREPADetect achieves higher speed and precision, making it an efficient sickle cell detection model suitable for resource-constrained clinical environments.

### IV. EXPERIMENTATION

#### A. Data Collection and Preprocessing

The experimental dataset consists of microscopic images from 20 SCD patients, collected at three sites: the UMRED Laboratory (University of Thiès), the Children's Hospital of Diamniadio (Senegal), and the Cell Environment Laboratory (Paris, France).

The 'SKDrepaData' [24] dataset, captured using an optical microscope at  $40\times$  and  $100\times$  magnifications, consists of 619 images. Annotation was performed using Labelme [25] with at least four hematology and biology experts, ensuring data quality and reliability. Images were categorized into Normal cells, Sickle cells, and other cell types, with their distinct morphological features illustrated in Fig 2. To enhance model generalization, variability and robustness, data augmentation techniques such as flipping,  $90^\circ$  rotations, random cropping (0–15%), and exposure adjustments (–20% to +20%) were applied. This increased the number of images to 1,487, which were split into 70% for training, 20% for validation, and 10% for testing.

#### B. Experimental Setup and Evaluation Metrics

Experiments were conducted with PyTorch 2.5.1 on an NVIDIA Tesla T4 GPU (15,102 MiB VRAM) with CUDA 12.1 for accelerated computations. Training spanned 200 epochs, with early stopping (patience = 100) to prevent overfitting while optimizing parameter exploration. A batch size of 12 balanced gradient stability and memory efficiency. The AdamW optimizer, known for its robust convergence, was used with an initial learning rate of 0.001, a decay factor of 0.1, and momentum of 0.9 to enhance stability.

Model performance was evaluated using Precision (P), Recall (R), mean Average Precision (mAP), and FLOPs [26] (equation 1). mAP combines precision and recall, with mAP@50 reflecting precision at an IoU of 0.5, and mAP@50-95 averaging precision across IoU thresholds (0.5–0.95, in 0.05 steps) for comprehensive evaluation. FLOPs quantify DREPADetect's computational complexity, measuring floating point operations per second, providing a key indicator of efficiency and processing speed.

$$P = \frac{TP}{TP + FP} ; R = \frac{TP}{TP + FN} ; mAP = \frac{1}{n} \sum_{i=1}^n AP_i \quad (1)$$

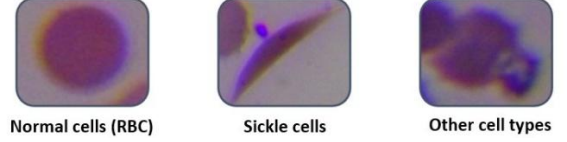


Fig 2. The three cell categories studied.

### V. RESULTS AND DISCUSSION

#### A. Comparative Analysis of Model Performance

DREPADetect's performance is evaluated during training and validation, highlighting the impact of its structural optimizations. For a fair comparison, models were tested on 'SKDrepaData' using identical hyperparameters. Performance metrics (precision, recall, mAP@50, mAP@50-95) are detailed in Fig. 3 and TABLE I.

As shown in Fig 3., after 25 epochs, DREPADetect consistently outperforms YOLOv8 in Precision and mAP, achieving  $P = 86.0\%$ ,  $mAP@50 = 90.0\%$ , and  $mAP@50-95 \approx 64.1\%$ , compared to YOLOv8's  $P = 84.5\%$ ,  $mAP@50 = 89.4\%$ , and  $mAP@50-95 \approx 63.7\%$ . However, YOLOv8 exhibits higher Recall ( $R = 88.6\%$ ), compared to DREPADetect's  $R = 87.6\%$ , indicating that while DREPADetect improves precision and detection accuracy, YOLOv8 captures a broader range of sickle cells, albeit with a higher false positive rate. The comparative validation table (Table I.) further confirms benefits of DREPADetect optimizations over the baseline.

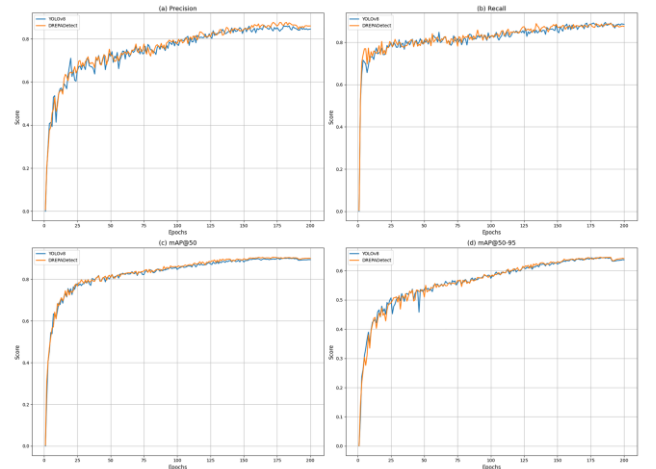


Fig 3. Comparison curves of P (a), R (b), mAP@50 (c) and mAP@50:95 (d) between models.

DREPADetect achieves 87.6% overall precision, outperforming YOLOv8's 84.6% by 3%. Additionally, it

exhibits superior mAP scores, with  $\text{mAP}@50 = 90.3\%$  and  $\text{mAP}@50-95 = 64.6\%$ , compared to YOLOv8's  $90.0\%$  and  $64.5\%$ , respectively.

After extensive validation, DREPADetect demonstrates once again its ability to identify each class individually and outperforms YOLOv8 in overall processing efficiency. Visual proof of these results is presented in Section **Erreur ! Source du renvoi introuvable.**

Considering hardware limitations in clinical environments, DREPADetect's computational efficiency is evaluated in detail in the next subsection.

DREPADetect outperforms YOLOv8 in sickle cell detection, achieving  $\text{mAP}@50$  of  $92.1\%$ , compared to  $91.6\%$  for YOLOv8. Similarly, for normal cells, DREPA-Detect reaches  $93.3\%$ , surpassing YOLOv8's  $92.4\%$ . However, for other cell types, YOLOv8 slightly outperforms DREPADetect, scoring  $86.1\%$  vs.  $85.4\%$ . These findings further validate the effectiveness of DREPADetect's structural optimizations, enhancing detection accuracy and robustness.

Table I. EXPERIMENTAL RESULTS OF ON THE VALIDATION DATASET

| Models      | Precision (%) | Recall (%) | $\text{mAP}@50$ (%) | $\text{mAP}@50:95$ (%) |
|-------------|---------------|------------|---------------------|------------------------|
| YOLOv8      | 84.6          | 88.8       | 90.0                | 64.5                   |
| DREPADetect | 87.6          | 87.6       | 90.3                | 64.6                   |

### B. Benchmarking Computational Efficiency

The experimental results on computational efficiency, measured in GFLOPs, high-light DREPADetect's superior resource management. The model significantly reduces computational complexity, requiring only 118.7 GFLOPs, compared to 164.8 GFLOPs for YOLOv8. Additionally, DREPADetect operates with 16,250,067 parameters, whereas YOLOv8 has 43,608,921, reflecting a  $62.7\%$  reduction in model size and a  $28\%$  decrease in computational demand for image processing.

The proposed optimizations also reduce the model's depth, with DREPADetect using 150 layers compared to 268 in YOLOv8. Additionally, image processing time improves by at least 4 milliseconds, with DREPADetect achieving 14.1ms per im-age, compared to 18.3ms for YOLOv8. These results confirm DREPADetect's efficiency, ensuring high performance while reducing computational costs, making it better suited for real-time clinical applications.

### C. Discussion

The experimental results confirm that DREPADetect, through architectural optimization, outperforms state-of-the-art models. Its structural innovations, specifically tailored to

this study, yielded highly satisfactory results across all evaluation stages.

During training, as shown in Fig 3., DREPADetect demonstrates a faster and more efficient learning process than YOLOv8. This improvement stems from the drastic reduction of non-essential layers, allowing the model to focus on crucial detection features, particularly cell size. Additionally, data preprocessing techniques—including augmentation—enhanced generalization capability, with these optimizations also reflected in validation performance.

In the validation phase, DREPADetect's optimized backbone and neck, designed for small object detection, enable the model to filter out irrelevant background elements and focus directly on target features. However, as shown in TABLE I, DREPA-Detect recorded a lower Recall ( $87.6\%$ ), likely due to the limited training data avail-able in this study. Recognizing the critical role of data quantity, we have made our annotated dataset publicly available to support further research in the scientific community. <https://doi.org/10.5281/zenodo.14906626>.

In cell identification and detection, DREPADetect enhances computational efficiency by  $28\%$  compared to YOLOv8, reducing operations per second while achieving  $22.9\%$  faster image processing. The model's simplified structure—150 layers versus 268 in YOLOv8—significantly boosts execution speed. These computational optimizations make DREPADetect highly suitable for clinical applications, where real-time processing and efficiency are essential.

### D. Experimental Comparative of Models on the SKDrepaData Dataset

DREPADetect demonstrates superior efficiency in sickle cell detection compared to recent models (TABLE II). While YOLOv7-tiny focuses on hypochromic microcytes (IDA) and AttriDet offers explainability but low performance ( $\text{mAP} = 28.2\%$ ), both lack clinical validation. YOLO-BC achieves high precision ( $80.9\%$  P,  $88.9\%$  R,  $\text{mAP}@50 = 90.1\%$ ) but is computationally intensive, hindering real-time use on low-end CPUs. YOLOv5-BS, optimized for speed, struggles with overlapping cells. Two stage models like R-CNN and Faster R-CNN provide  $>90\%$  precision but suffer from slow inference and poor generalization. Custom Deep CNNs are noise-sensitive, and CST-YOLO, despite high recall ( $92.7\%$ ), is computationally demanding and untested on specific diseases.

DREPADetect strikes an optimal balance, achieving  $87.6\%$  precision,  $87.6\%$  recall, and  $90.3\%$   $\text{mAP}@50$ . Tailored for sickle cell detection, it ensures real-time feasibility and suitability for low resource clinical settings. This study also introduces a new expert annotated dataset, enhancing reproducibility and clinical validation.

Table II. COMPARATIVE EXPERIMENT WITH STAT-OF-THE-ART MODELS

| Models           | Datasets (Classes) | P (%) | R (%) | $\text{mAP}@50$ | Observations                                                         |
|------------------|--------------------|-------|-------|-----------------|----------------------------------------------------------------------|
| YOLOv7-tiny [16] | Private (2)        | N/A   | N/A   | 89.0            | Specialized in hypochromic microcytes (IDA). No clinical validation. |

|                      |                          |             |             |             |                                                                                |
|----------------------|--------------------------|-------------|-------------|-------------|--------------------------------------------------------------------------------|
| AttriDet [14]        | <i>Private (14)</i>      | N/A         | N/A         | 28.2        | High explainability - low performance. High computational consumption.         |
| YOLO-BC [22]         | <i>Public (3)</i>        | 80.9        | 88.9        | 90.1        | High precision and computational load. Unsuitable for real-time practice       |
| YOLOv5-BS [17]       | <i>Public (3)</i>        | 92.8        | 83.8        | N/A         | Optimized for speed but limited handling of overlapping cells.                 |
| R-CNN [19]           | <i>Private (1)</i>       | 91.0        | N/A         | N/A         | Slow model and limited dataset. Poor generalization.                           |
| Faster R-CNN [20]    | <i>Public (3)</i>        | 90.0        | N/A         | N/A         | Good precision but high inference time. Not optimized for specific cell types. |
| Custom Deep CNN [13] | <i>Private (3)</i>       | 81.0        | N/A         | N/A         | Sensitive to noise and overlapping cells. High computational load.             |
| CST-YOLO [15]        | <i>Public (3)</i>        | N/A         | 92.7        | N/A         | High recall - very computationally intensive. Not tested on specific diseases. |
| <b>DREPADetect</b>   | <b><i>Public (3)</i></b> | <b>87.6</b> | <b>87.6</b> | <b>90.3</b> | <b>Good accuracy/speed balance. Suitable for low-resource environments.</b>    |

#### E. Visual evaluation of DREPADetect on Test Data

Fig 4. evaluates DREPADetect robustness across different conditions using test datasets images of both ‘SKDrepaData’ and ‘Erythrocyte IBD1’ [28]. In row (a), corresponding to ‘SKDrepaData’ images at 100× magnification, DREPADetect accurately detects well separated cells, correctly regressing bounding boxes with no missed detections. The model exhibits strong resilience to morphological variations and image noise, confidently classifying partially atypical cells into the appropriate ‘Other’ classes. In row (b), using ‘Erythrocyte IBD1’ images at 40× magnification, the detection scenario becomes more challenging due to overlapping cells, blurriness, and occlusions. Despite smaller cell sizes, DREPADetect successfully detects and classifies the cells, demonstrating adaptability to complex imaging conditions. These findings highlight DREPADetect’s robustness in handling size variations, cell overlaps, and visual artifacts, further reinforcing its potential for clinical applications across diverse image quality levels and scales.

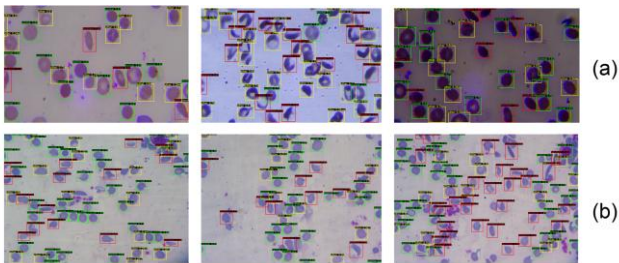


Fig 4. Visual discrimination of RBC Detection on SKDrepaData (a) and Erythrocyte IBD1(b) Datasets

#### VI. CONCLUSION AND FUTURE PERSPECTIVES

We developed DREPADetect, an optimized YOLOv8-based method for blood cell detection and counting. Tailored for clinical use, it detects small-sized cells (normal, sickle, and

others) while ensuring real-time performance for practical applications. Experimental results demonstrate that DREPADetect outperforms YOLOv8, achieving a precision of 87.6%, mAP@50 of 90.3%, and mAP@50-95 of 64.6%. Additionally, the model significantly reduces computational complexity, requiring 118.7 GFLOPs and 150 layers, compared to 164.8 GFLOPs and 268 layers for YOLOv8, thereby improving speed and efficiency while maintaining superior performance. A comparative analysis with other models, such as recent YOLO derivatives propositions and Faster R-CNN, confirms that DREPADetect delivers better overall performance in red blood cell detection, surpassing these models in terms of precision, recall, and mAP.

Our future research will focus on: (1) Developing an integrated DREPADetect based system for sickle cell detection and counting, ensuring accessible diagnostics in remote areas of Senegal. (2) Expanding and releasing a public sickle cell dataset to mitigate data scarcity and advance hematological research. (3) Exploring federated learning to facilitate secure collaboration across medical centers, improving model performance and clinical adoption.

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